
```
1  # Says hello to world by returning a string of text
2
3  from flask import Flask
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return "hello, world"
```

```
1  # Says hello to world by returning a string of HTML
2
3  from flask import Flask
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return '<!DOCTYPE html><html lang="en"><head><title>hello</title></head><body>hello, world</body></html>'
```

```
1  # Says hello to world
2
3  from flask import Flask, render_template
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return render_template("index.html")
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         hello, world
11     </body>
12
13 </html>
```

```
1  # Says hello to request.args["name"]
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     if "name" in request.args:
11         name = request.args["name"]
12     else:
13         name = "world"
14     return render_template("index.html", placeholder=name)
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>hello</title>
8      </head>
9
10     <body>
11         hello, {{ placeholder }}
12     </body>
13
14 </html>
```

```
1  # Uses parameter with same name as variable
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     if "name" in request.args:
11         name = request.args["name"]
12     else:
13         name = "world"
14     return render_template("index.html", name=name)
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         hello, {{ name }}
11     </body>
12
13 </html>
```



```
1  # Uses request.args.get
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     name = request.args.get("name", "world")
11     return render_template("index.html", name=name)
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         hello, {{ name }}
11     </body>
12
13 </html>
```

```
1  # Adds a form, second route
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return render_template("index.html")
11
12
13  @app.route("/greet")
14  def greet():
15     return render_template("greet.html", name=request.args.get("name", "world"))
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>hello</title>
8      </head>
9
10     <body>
11         hello, {{ name }}
12     </body>
13
14 </html>
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         <form action="/greet" method="get">
11             <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
12             <button type="submit">Greet</button>
13         </form>
14     </body>
15
16 </html>
```

```
1  # Adds a layout
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return render_template("index.html")
11
12
13  @app.route("/greet")
14  def greet():
15     return render_template("greet.html", name=request.args.get("name", "world"))
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     hello, {{ name }}
5 {% endblock %}
```

```
1  {% extends "layout.html" %}
2
3  {% block body %}
4
5      <form action="/greet" method="get">
6          <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7          <button type="submit">Greet</button>
8      </form>
9
10 {% endblock %}
```



```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         {% block body %}{% endblock %}
11     </body>
12
13 </html>
```

```
1  # Switches to POST
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return render_template("index.html")
11
12
13  @app.route("/greet", methods=["POST"])
14  def greet():
15     return render_template("greet.html", name=request.form.get("name", "world"))
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     hello, {{ name }}
5 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <form action="/greet" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         <button type="submit">Greet</button>
8     </form>
9
10 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         {% block body %}{% endblock %}
11     </body>
12
13 </html>
```

```
1  # Uses a single route
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/", methods=["GET", "POST"])
9  def index():
10     if request.method == "POST":
11         return render_template("greet.html", name=request.form.get("name", "world"))
12     return render_template("index.html")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     hello, {{ name }}
6
7 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <form action="/" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         <button type="submit">Greet</button>
8     </form>
9
10 {% endblock %}
```



```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         {% block body %}{% endblock %}
11     </body>
12
13 </html>
```

```
1  # Moves default value to template
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/", methods=["GET", "POST"])
9  def index():
10     if request.method == "POST":
11         return render_template("greet.html", name=request.form.get("name"))
12     return render_template("index.html")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     hello,
6     {% if name %}
7         {{ name }}
8     {% else %}
9         world
10    {% endif %}
11
12 {% endblock %}
```

```
1  {% extends "layout.html" %}
2
3  {% block body %}
4
5      <form action="/" method="post">
6          <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7          <button type="submit">Greet</button>
8      </form>
9
10 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <title>hello</title>
7      </head>
8
9      <body>
10         {% block body %}{% endblock %}
11     </body>
12
13 </html>
```

```
1  # Implements a registration form using a select menu without validating sport server-side
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7
8  @app.route("/")
9  def index():
10     return render_template("index.html")
11
12
13  @app.route("/register", methods=["POST"])
14  def register():
15
16     # Validate submission
17     if not request.form.get("name") or not request.form.get("sport"):
18         return render_template("failure.html")
19
20     # Confirm registration
21     return render_template("success.html")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are not registered!
5 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         <select name="sport">
8             <option selected value="">Sport</option>
9             <option value="Basketball">Basketball</option>
10            <option value="Soccer">Soccer</option>
11            <option value="Ultimate Frisbee">Ultimate Frisbee</option>
12        </select>
13        <button type="submit">Register</button>
14    </form>
15 {% endblock %}
```



```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are registered!
5 {% endblock %}
```

```
1  # Implements a registration form using a select menu, validating sport server-side
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7  SPORTS = [
8      "Basketball",
9      "Soccer",
10     "Ultimate Frisbee"
11 ]
12
13
14 @app.route("/")
15 def index():
16     return render_template("index.html", sports=SPORTS)
17
18
19 @app.route("/register", methods=["POST"])
20 def register():
21
22     # Validate submission
23     if not request.form.get("name") or request.form.get("sport") not in SPORTS:
24         return render_template("failure.html")
25
26     # Confirm registration
27     return render_template("success.html")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are not registered!
5 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         <select name="sport">
8             <option selected value="">Sport</option>
9             {% for sport in sports %}
10                 <option value="{{ sport }}">{{ sport }}</option>
11             {% endfor %}
12         </select>
13         <button type="submit">Register</button>
14     </form>
15 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are registered!
5 {% endblock %}
```

```
1  # Implements a registration form using radio buttons
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7  SPORTS = [
8      "Basketball",
9      "Soccer",
10     "Ultimate Frisbee"
11 ]
12
13
14 @app.route("/")
15 def index():
16     return render_template("index.html", sports=SPORTS)
17
18
19 @app.route("/register", methods=["POST"])
20 def register():
21
22     # Validate submission
23     if not request.form.get("name") or request.form.get("sport") not in SPORTS:
24         return render_template("failure.html")
25
26     # Confirm registration
27     return render_template("success.html")
```



```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are not registered!
5 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="radio" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are registered!
5 {% endblock %}
```

```
1  # Adds error messages
2
3  from flask import Flask, render_template, request
4
5  app = Flask(__name__)
6
7  SPORTS = [
8      "Basketball",
9      "Soccer",
10     "Ultimate Frisbee"
11 ]
12
13
14 @app.route("/")
15 def index():
16     return render_template("index.html", sports=SPORTS)
17
18
19 @app.route("/register", methods=["POST"])
20 def register():
21
22     # Validate name
23     name = request.form.get("name")
24     if not name:
25         return render_template("error.html", message="Missing name")
26
27     # Validate sport
28     sport = request.form.get("sport")
29     if not sport:
30         return render_template("error.html", message="Missing sport")
31     if sport not in SPORTS:
32         return render_template("error.html", message="Invalid sport")
33
34     # Confirm registration
35     return render_template("success.html")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Error</h1>
5     <p>{{ message }}</p>
6     
7 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="radio" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```



```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     You are registered!
5 {% endblock %}
```

```
1  # Stores registrants in a dictionary
2
3  from flask import Flask, redirect, render_template, request
4
5  app = Flask(__name__)
6
7  REGISTRANTS = {}
8
9  SPORTS = [
10     "Basketball",
11     "Soccer",
12     "Ultimate Frisbee"
13 ]
14
15
16 @app.route("/")
17 def index():
18     return render_template("index.html", sports=SPORTS)
19
20
21 @app.route("/register", methods=["POST"])
22 def register():
23
24     # Validate name
25     name = request.form.get("name")
26     if not name:
27         return render_template("error.html", message="Missing name")
28
29     # Validate sport
30     sport = request.form.get("sport")
31     if not sport:
32         return render_template("error.html", message="Missing sport")
33     if sport not in SPORTS:
34         return render_template("error.html", message="Invalid sport")
35
36     # Remember registrant
37     REGISTRANTS[name] = sport
38
39     # Confirm registration
40     return redirect("/registrants")
41
42
```

```
43 @app.route("/registrants")
44 def registrants():
45     return render_template("registrants.html", registrants=REGISTRANTS)
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Error</h1>
5     <p>{{ message }}</p>
6     
7 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="radio" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1  {% extends "layout.html" %}
2
3  {% block body %}
4      <h1>Registrants</h1>
5      <table>
6          <thead>
7              <tr>
8                  <th>Name</th>
9                  <th>Sport</th>
10             </tr>
11         </thead>
12         <tbody>
13             {% for name in registrants %}
14                 <tr>
15                     <td>{{ name }}</td>
16                     <td>{{ registrants[name] }}</td>
17                 </tr>
18             {% endfor %}
19         </tbody>
20     </table>
21 {% endblock %}
```

```
1  # Stores registrants in a SQLite database
2
3  from cs50 import SQL
4  from flask import Flask, redirect, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///froshims.db")
9
10 SPORTS = [
11     "Basketball",
12     "Soccer",
13     "Ultimate Frisbee"
14 ]
15
16
17 @app.route("/")
18 def index():
19     return render_template("index.html", sports=SPORTS)
20
21
22 @app.route("/register", methods=["POST"])
23 def register():
24
25     # Validate name
26     name = request.form.get("name")
27     if not name:
28         return render_template("error.html", message="Missing name")
29
30     # Validate sport
31     sport = request.form.get("sport")
32     if not sport:
33         return render_template("error.html", message="Missing sport")
34     if sport not in SPORTS:
35         return render_template("error.html", message="Invalid sport")
36
37     # Remember registrant
38     db.execute("INSERT INTO registrants (name, sport) VALUES(?, ?)", name, sport)
39
40     # Confirm registration
41     return redirect("/registrants")
42
```

```
43
44 @app.route("/registrants")
45 def registrants():
46     registrants = db.execute("SELECT * FROM registrants")
47     return render_template("registrants.html", registrants=registrants)
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Error</h1>
5     <p>{{ message }}</p>
6     
7 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="radio" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1  {% extends "layout.html" %}
2
3  {% block body %}
4      <h1>Registrants</h1>
5      <table>
6          <thead>
7              <tr>
8                  <th>Name</th>
9                  <th>Sport</th>
10                 <th></th>
11             </tr>
12         </thead>
13         <tbody>
14             {% for registrant in registrants %}
15                 <tr>
16                     <td>{{ registrant["name"] }}</td>
17                     <td>{{ registrant["sport"] }}</td>
18                 </tr>
19             {% endfor %}
20         </tbody>
21     </table>
22 {% endblock %}
```

```
1  # Implements a registration form, storing registrants in a SQLite database, with support for deregistration
2
3  from cs50 import SQL
4  from flask import Flask, redirect, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///froshims.db")
9
10 SPORTS = [
11     "Basketball",
12     "Soccer",
13     "Ultimate Frisbee"
14 ]
15
16
17 @app.route("/")
18 def index():
19     return render_template("index.html", sports=SPORTS)
20
21
22 @app.route("/deregister", methods=["POST"])
23 def deregister():
24
25     # Forget registrant
26     id = request.form.get("id")
27     if id:
28         db.execute("DELETE FROM registrants WHERE id = ?", id)
29     return redirect("/registrants")
30
31
32 @app.route("/register", methods=["POST"])
33 def register():
34
35     # Validate name
36     name = request.form.get("name")
37     if not name:
38         return render_template("error.html", message="Missing name")
39
40     # Validate sport
41     sport = request.form.get("sport")
42     if not sport:
```

```
43     return render_template("error.html", message="Missing sport")
44 if sport not in SPORTS:
45     return render_template("error.html", message="Invalid sport")
46
47 # Remember registrant
48 db.execute("INSERT INTO registrants (name, sport) VALUES(?, ?)", name, sport)
49
50 # Confirm registration
51 return redirect("/registrants")
52
53
54 @app.route("/registrants")
55 def registrants():
56     registrants = db.execute("SELECT * FROM registrants")
57     return render_template("registrants.html", registrants=registrants)
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Error</h1>
5     <p>{{ message }}</p>
6     
7 {% endblock %}
```



```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="radio" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Registrants</h1>
5     <table>
6         <thead>
7             <tr>
8                 <th>Name</th>
9                 <th>Sport</th>
10                <th></th>
11            </tr>
12        </thead>
13        <tbody>
14            {% for registrant in registrants %}
15                <tr>
16                    <td>{{ registrant["name"] }}</td>
17                    <td>{{ registrant["sport"] }}</td>
18                    <td>
19                        <form action="/deregister" method="post">
20                            <input name="id" type="hidden" value="{{ registrant.id }}">
21                            <button type="submit">Deregister</button>
22                        </form>
23                    </td>
24                </tr>
25            {% endfor %}
26        </tbody>
27    </table>
28 {% endblock %}
```

```
1  # Implements a registration form, storing registrants in a SQLite database, with support for multiple sports and  
deregistration  
2  
3  from cs50 import SQL  
4  from flask import Flask, redirect, render_template, request  
5  
6  app = Flask(__name__)  
7  
8  db = SQL("sqlite:///froshims.db")  
9  
10 SPORTS = [  
11     "Basketball",  
12     "Soccer",  
13     "Ultimate Frisbee"  
14 ]  
15  
16  
17 @app.route("/")  
18 def index():  
19     return render_template("index.html", sports=SPORTS)  
20  
21  
22 @app.route("/deregister", methods=["POST"])  
23 def deregister():  
24  
25     # Forget registrant  
26     id = request.form.get("id")  
27     if id:  
28         db.execute("DELETE FROM registrants WHERE id = ?", id)  
29     return redirect("/registrants")  
30  
31  
32 @app.route("/register", methods=["POST"])  
33 def register():  
34  
35     # Validate name  
36     name = request.form.get("name")  
37     if not name:  
38         return render_template("error.html", message="Missing name")  
39  
40     # Validate sports  
41     sports = request.form.getlist("sport")
```

```
42     if not sports:
43         return render_template("error.html", message="Missing sport")
44     for sport in sports:
45         if sport not in SPORTS:
46             return render_template("error.html", message="Invalid sport")
47
48     # Remember registrant
49     for sport in sports:
50         db.execute("INSERT INTO registrants (name, sport) VALUES(?, ?)", name, sport)
51
52     # Confirm registration
53     return redirect("/registrants")
54
55
56 @app.route("/registrants")
57 def registrants():
58     registrants = db.execute("SELECT * FROM registrants")
59     return render_template("registrants.html", registrants=registrants)
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Error</h1>
5     <p>{{ message }}</p>
6     
7 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Register</h1>
5     <form action="/register" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         {% for sport in sports %}
8             <input name="sport" type="checkbox" value="{{ sport }}"> {{ sport }}
9         {% endfor %}
10        <button type="submit">Register</button>
11    </form>
12 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>froshims</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```



```
1 {% extends "layout.html" %}
2
3 {% block body %}
4     <h1>Registrants</h1>
5     <table>
6         <thead>
7             <tr>
8                 <th>Name</th>
9                 <th>Sport</th>
10                <th></th>
11            </tr>
12        </thead>
13        <tbody>
14            {% for registrant in registrants %}
15                <tr>
16                    <td>{{ registrant["name"] }}</td>
17                    <td>{{ registrant["sport"] }}</td>
18                    <td>
19                        <form action="/deregister" method="post">
20                            <input name="id" type="hidden" value="{{ registrant.id }}">
21                            <button type="submit">Deregister</button>
22                        </form>
23                    </td>
24                </tr>
25            {% endfor %}
26        </tbody>
27    </table>
28 {% endblock %}
```

```
1  from flask import Flask, redirect, render_template, request, session
2  from flask_session import Session
3
4  # Configure app
5  app = Flask(__name__)
6
7  # Configure session
8  app.config["SESSION_PERMANENT"] = False
9  app.config["SESSION_TYPE"] = "filesystem"
10 Session(app)
11
12
13 @app.route("/")
14 def index():
15     return render_template("index.html", name=session.get("name"))
16
17
18 @app.route("/login", methods=["GET", "POST"])
19 def login():
20     if request.method == "POST":
21         session["name"] = request.form.get("name")
22         return redirect("/")
23     return render_template("login.html")
24
25
26 @app.route("/logout")
27 def logout():
28     session.clear()
29     return redirect("/")
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     {% if name %}
6         You are logged in as {{ name }}. <a href="/logout">Log out</a>.
7     {% else %}
8         You are not logged in. <a href="/login">Log in</a>.
9     {% endif %}
10
11 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>login</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <form action="/login" method="post">
6         <input autocomplete="off" autofocus name="name" placeholder="Name" type="text">
7         <button type="submit">Log In</button>
8     </form>
9
10 {% endblock %}
```

```
1  from cs50 import SQL
2  from flask import Flask, redirect, render_template, request, session
3  from flask_session import Session
4
5  # Configure app
6  app = Flask(__name__)
7
8  # Connect to database
9  db = SQL("sqlite:///store.db")
10
11 # Configure session
12 app.config["SESSION_PERMANENT"] = False
13 app.config["SESSION_TYPE"] = "filesystem"
14 Session(app)
15
16
17 @app.route("/")
18 def index():
19     books = db.execute("SELECT * FROM books")
20     return render_template("books.html", books=books)
21
22
23 @app.route("/cart", methods=["GET", "POST"])
24 def cart():
25
26     # Ensure cart exists
27     if "cart" not in session:
28         session["cart"] = []
29
30     # POST
31     if request.method == "POST":
32         book_id = request.form.get("id")
33         if book_id:
34             session["cart"].append(book_id)
35             return redirect("/cart")
36
37     # GET
38     books = db.execute("SELECT * FROM books WHERE id IN (?)", session["cart"])
39     return render_template("cart.html", books=books)
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <h1>Books</h1>
6     {% for book in books %}
7         <h2>{{ book["title"] }}</h2>
8         <form action="/cart" method="post">
9             <input name="id" type="hidden" value="{{ book['id'] }}">
10             <button type="submit">Add to Cart</button>
11         </form>
12     {% endfor %}
13
14 {% endblock %}
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <h1>Cart</h1>
6     <ol>
7         {% for book in books %}
8             <li>{{ book["title"] }}</li>
9         {% endfor %}
10    </ol>
11
12 {% endblock %}
```



```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>store</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1  # Searches for shows
2
3  from cs50 import SQL
4  from flask import Flask, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///shows.db")
9
10
11 @app.route("/")
12 def index():
13     return render_template("index.html")
14
15
16 @app.route("/search")
17 def search():
18     shows = db.execute("SELECT * FROM shows WHERE title = ?", request.args.get("q"))
19     return render_template("search.html", shows=shows)
```

1 <https://www.imdb.com/conditions>

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <form action="/search" method="get">
6         <input autocomplete="off" autofocus name="q" placeholder="Query" type="search">
7         <button type="submit">Search</button>
8     </form>
9
10 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>shows</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <ul>
6         {% for show in shows %}
7             <li>{{ show["title"] }}</li>
8         {% endfor %}
9     </ul>
10
11 {% endblock %}
```

```
1  # Searches for shows using LIKE
2
3  from cs50 import SQL
4  from flask import Flask, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///shows.db")
9
10
11 @app.route("/")
12 def index():
13     return render_template("index.html")
14
15
16 @app.route("/search")
17 def search():
18     shows = db.execute("SELECT * FROM shows WHERE title LIKE ?", "%" + request.args.get("q") + "%")
19     return render_template("search.html", shows=shows)
```

1 <https://www.imdb.com/conditions>


```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <form action="/search" method="get">
6         <input autocomplete="off" autofocus name="q" placeholder="Query" type="search">
7         <button type="submit">Search</button>
8     </form>
9
10 {% endblock %}
```

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>shows</title>
8      </head>
9
10     <body>
11         {% block body %}{% endblock %}
12     </body>
13
14 </html>
```

```
1 {% extends "layout.html" %}
2
3 {% block body %}
4
5     <ul>
6         {% for show in shows %}
7             <li>{{ show["title"] }}</li>
8         {% endfor %}
9     </ul>
10
11 {% endblock %}
```

```
1  # Searches for shows using Ajax
2
3  from cs50 import SQL
4  from flask import Flask, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///shows.db")
9
10
11 @app.route("/")
12 def index():
13     return render_template("index.html")
14
15
16 @app.route("/search")
17 def search():
18     q = request.args.get("q")
19     if q:
20         shows = db.execute("SELECT * FROM shows WHERE title LIKE ? LIMIT 50", "%" + q + "%")
21     else:
22         shows = []
23     return render_template("search.html", shows=shows)
```

1 <https://www.imdb.com/conditions>

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>shows</title>
8      </head>
9
10     <body>
11
12         <input autocomplete="off" autofocus placeholder="Query" type="search">
13
14         <ul></ul>
15
16         <script>
17             let input = document.querySelector('input');
18             input.addEventListener('input', async function() {
19                 let response = await fetch('/search?q=' + input.value);
20                 let shows = await response.text();
21                 document.querySelector('ul').innerHTML = shows;
22             });
23         </script>
24
25     </body>
26
27 </html>
```

```
1 {% for show in shows %}
2     <li>{{ show["title"] }}</li>
3 {% endfor %}
```

```
1  # Searches for shows using Ajax with JSON
2
3  from cs50 import SQL
4  from flask import Flask, jsonify, render_template, request
5
6  app = Flask(__name__)
7
8  db = SQL("sqlite:///shows.db")
9
10
11 @app.route("/")
12 def index():
13     return render_template("index.html")
14
15
16 @app.route("/search")
17 def search():
18     q = request.args.get("q")
19     if q:
20         shows = db.execute("SELECT * FROM shows WHERE title LIKE ? LIMIT 50", "%" + q + "%")
21     else:
22         shows = []
23     return jsonify(shows)
```

1 <https://www.imdb.com/conditions>

```
1  <!DOCTYPE html>
2
3  <html lang="en">
4
5      <head>
6          <meta name="viewport" content="initial-scale=1, width=device-width">
7          <title>shows</title>
8      </head>
9
10     <body>
11
12         <input autocomplete="off" autofocus placeholder="Query" type="text">
13
14         <ul></ul>
15
16         <script>
17             let input = document.querySelector('input');
18             input.addEventListener('input', async function() {
19                 let response = await fetch('/search?q=' + input.value);
20                 let shows = await response.json();
21                 let html = '';
22                 for (let i in shows) {
23                     let title = shows[i].title.replace('<', '&lt;').replace('&', '&amp;');
24                     html += '<li>' + title + '</li>';
25                 }
26                 document.querySelector('ul').innerHTML = html;
27             });
28         </script>
29
30     </body>
31
32 </html>
```